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Experiment 2: To Containerize Application using Docker

Aim:

To containerize application using Docker .

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory:

Steps to install docker: i.

Open your preferred web browser and visit the official Docker website at **https://www.docker.com/**. Download the latest version of **Docker Desktop** that is compatible with the Windows operating system.

ii.

After the download is complete, locate the installer in your Downloads folder (or the location where it was saved) and launch it by double-clicking the executable file.

iii.

Proceed through the installation wizard by following the displayed instructions. During the setup, ensure that the **"Use WSL 2 instead of Hyper-V"** option is enabled, as it offers improved compatibility and performance for Windows users.

iv.

Allow the installation process to complete successfully. Once finished, click **"Close"**. If prompted, restart your computer to finalize the installation.

v.

After the system restarts, open the **Start Menu**, search for **Docker Desktop**, and launch the application. If prompted, review and accept the license agreement so that Docker Desktop can initialize the Docker Engine.

vi.

Open **Command Prompt**, **PowerShell**, or **Windows Terminal**, and execute the following command:

Docker: Docker is an open-source containerization platform designed to simplify software development, testing, and deployment. It enables developers to package an application together with all its required libraries, dependencies, runtime components, and configuration files into a single portable unit. This approach guarantees that the application behaves consistently across different computing environments, including development systems, testing servers, and production infrastructure. By eliminating dependency and configuration differences between systems, Docker helps avoid compatibility issues commonly described as *"it works on my machine."*

Docker Images: A Docker image is a pre-built, lightweight, and immutable template that contains everything needed to execute an application. It serves as the blueprint from which Docker containers are created.

Internally, an image consists of multiple read-only layers, where each layer represents a specific build instruction such as installing software packages, configuring environment variables, copying application files, or setting execution commands. Along with the application code, a Docker image includes the runtime environment, operating system libraries, required system utilities, and default configuration settings. Since images are immutable, their contents cannot be modified after they have been created. Any changes require building a new image.

Docker Containers: A Docker container is a running instance of a Docker image. It provides an isolated execution environment in which the packaged application operates independently from both the host operating system and other containers.

When a container is launched, Docker creates a thin writable layer on top of the image's read-only layers. This writable layer temporarily stores runtime changes such as generated files, application logs, and user data while the container is active. The underlying image remains unchanged, ensuring consistency across deployments. Container isolation improves security, portability, and reliability, allowing applications to execute in the same manner regardless of the hardware or operating system on which they are deployed.

Implementation: 1. Verify Docker installation.

```
docker --version
```

2. Search for the required Docker image.

```
docker search mysql
```

3. Download the MySQL image.

```
docker pull mysql
```

4. Create and run the MySQL container.

```
docker run -d --name lab_db -e MYSQL_ROOT_PASSWORD=secret mysql
```

5. Connect to the MySQL container.

docker

docker exec -it lab_db mysql -u root -p

6. Create a new database.

```
CREATE DATABASE lab_workexp2;
```

7. Select the database.

```
USE lab_workexp2;
```

8. Create a table.

```
CREATE TABLE stud (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50),
    marks INT
);
```

9. Insert a record into the table.

```
INSERT INTO stud (roll_no, name, marks)
VALUES (1, 'Raj', 95);
```

10. Display the inserted record.

```
SELECT * FROM stud
```

```
C:\Users\15L>docker search ubuntu
NAME                              DESCRIPTION                                     STARS     OFFICIAL
ubuntu                            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                      Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                      BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                      Apache Kafka, a distributed event streaming …   62
ubuntu/apache2                    Apache, a secure & extensible open-source HT…   104
ubuntu/zookeeper                  ZooKeeper maintains configuration informatio…   15
ubuntu/prometheus                 Prometheus is a systems and service monitori…   80
ubuntu/jre                        Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet              Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                     A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                   PostgreSQL is an open source object-relation…   45
ubuntu/redis                      Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-runtime             Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps                Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                    Grafana, a feature rich metrics dashboard & …   12
ubuntu/jdk                        Ubuntu rock for OpenJDK                         0
ubuntu/memcached                  Memcached, in-memory keyvalue store for smal…   7
ubuntu/rabbitmq                   Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                      Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                       Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                    Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/alertmanager               Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/opentelemetry-collector    This is a rock for the OpenTelemetry Collect…   0
```

1. search ubuntu --filter stars=100

```
C:\Users\15L>docker search ubuntu --filter stars=100
NAME              DESCRIPTION                                     STARS     OFFICIAL
ubuntu            Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid      Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx      Nginx, a high-performance reverse proxy & we…   141
ubuntu/apache2    Apache, a secure & extensible open-source HT…   104
ubuntu/bind9      BIND 9 is a very flexible, full-featured DNS…   120
```

2. docker search redis

```
C:\Users\15L>docker search redis
NAME                              DESCRIPTION                                     STARS     OFFICIAL
redis                             Redis is the world's fastest data platform f…   13604     [OK]
redis/redis-stack-server          redis-stack-server installs a Redis server w…   104
redis/redis-stack                 redis-stack installs a Redis server with add…   162
redis/redisinsight                Redis Insight - our best official GUI for Re…   54
redislabs/redis                   Clustered in-memory database engine compatib…   46
redis/rdi-operator                                                                0
redis/rdi-collector-initializer    Init container for RDI Collector               0
redis/rdi-api                                                                     0
redis/rdi-monitor                                                                 0
redis/rdi-collector-api                                                           0
redis/rdi-processor                                                               1
redis/rdi-cli                                                                     0
redis/rdi-flink-processor                                                         0
redis/reloader                                                                    0
redis/rdi-metrics-aggregator                                                      0
redis/rdi-flink-collector                                                         0
circleci/redis                    CircleCI images for Redis                       18
redis/sidekiq-benchmark           Sidekiq load benchmark in Rust - throughput …   0
redis/redis-benchmark-go          redis-benchmark utility in go.                  1
redis/vector-db-benchmark         Framework for benchmarking vector search eng…   1
bitnamicharts/redis               Bitnami Helm chart for Redis(R)                 37
redis/bcast-tracking-bench        Redis CLIENT TRACKING BCAST invalidation lis…   0
redis/redis-di-cli                                                                0
redis/pubsub-sub-bench                                                            0
cimg/redis                                                                        2
```

3. docker pull redis

```
C:\Users\15L>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
fd6b3803006b: Pull complete
6f0ec8fc933c: Pull complete
c1af3958d6f8: Pull complete
234dd6735104: Pull complete
4f4fb700ef54: Pull complete
8fd80e87c6f1: Pull complete
280ce3063e4e: Download complete
1bd39139fcfd: Download complete
Digest: sha256:234c902a2db49461a129e2d4aeff85b28cf20187ed274a67f6e50995fa713c7b
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest
```

4. docker run -d --name my_database redis

```
C:\Users\15L>docker run -d --name my_database redis
f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7
```

5. docker ps

docker

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
f7683dee4227   redis     "docker-entrypoint.s…"   9 seconds ago   Up 8 seconds   6379/tcp                                my_database
63f5d6f4f6d8   nginx     "/docker-entrypoint.…"   7 minutes ago   Up 7 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp   nginx

C:\Users\15L>docker inspect my_database
```

6. docker inspect my_database


```
C:\Users\15L>docker inspect my_database
[
    {
        "Id": "f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7",
        "Created": "2026-07-22T08:19:23.942616168Z",
        "Path": "docker-entrypoint.sh",
        "Args": [
            "redis-server"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 818,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-07-22T08:19:23.9956475433Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:234c902a2db49461a129e2d4aeff85b28cf20187ed274a67f6e50995fa713c7b",
        "ResolvConfPath": "/var/lib/docker/containers/f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7/resolv.conf
        "HostnamePath": "/var/lib/docker/containers/f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7/hostname",
        "HostsPath": "/var/lib/docker/containers/f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7/hosts",
        "LogPath": "/var/lib/docker/containers/f7683dee42274a949b10471949f259cbcdbdb7c4e7e782db36debb48ce51dbd7/f7683dee42274a949b
        "Name": "/my_database",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
```


7. docker logs -f my_database


```
C:\Users\15L>docker logs -f my_database
Starting Redis Server
1:C 22 Jul 2026 08:19:24.138 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 22 Jul 2026 08:19:24.138 * Redis version=8.8.0, bits=64, commit=00000000, modified=1, pid=1, just started
1:C 22 Jul 2026 08:19:24.138 * Configuration loaded
1:M 22 Jul 2026 08:19:24.138 * monotonic clock: POSIX clock_gettime
1:M 22 Jul 2026 08:19:24.139 * Running mode=standalone, port=6379.
1:M 22 Jul 2026 08:19:24.139 * <bf> RedisBloom version 8.8.0 (Git=unknown)
1:M 22 Jul 2026 08:19:24.139 * <bf> Registering configuration options: [
1:M 22 Jul 2026 08:19:24.139 * <bf>     { bf-error-rate       :       0.01 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { bf-initial-size     :        100 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { bf-expansion-factor :          2 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { cf-bucket-size      :          2 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { cf-initial-size     :       1024 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { cf-max-iterations   :         20 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { cf-expansion-factor :          1 }
1:M 22 Jul 2026 08:19:24.139 * <bf>     { cf-max-expansions   :         32 }
1:M 22 Jul 2026 08:19:24.139 * <bf> ]
1:M 22 Jul 2026 08:19:24.139 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
1:M 22 Jul 2026 08:19:24.146 * <search> search-workers default: 16 (min of MAX_WORKER_THREADS=16 and CPU cores)
1:M 22 Jul 2026 08:19:24.146 * <search> Redis version found by RedisSearch : 8.8.0 - oss
1:M 22 Jul 2026 08:19:24.146 * <search> RediSearch version 8.8.0 (Git=d2ae026)
1:M 22 Jul 2026 08:19:24.146 * <search> Low level api version 1 initialized successfully
1:M 22 Jul 2026 08:19:24.146 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read si
ze: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
1:M 22 Jul 2026 08:19:24.146 * <search> Initialized thread pools!
1:M 22 Jul 2026 08:19:24.146 * <search> Enabled workers threadpool of size 16
1:M 22 Jul 2026 08:19:24.148 * <search> Subscribe to config changes
1:M 22 Jul 2026 08:19:24.148 * <search> Subscribe to cluster slot migration events
1:M 22 Jul 2026 08:19:24.148 * <search> Enabled role change notification
1:M 22 Jul 2026 08:19:24.149 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
1:M 22 Jul 2026 08:19:24.149 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
1:M 22 Jul 2026 08:19:24.149 * <timeseries> RedisTimeSeries version 80800, git_sha=42ca4f1078fca732b7f9256adbf25914d67e1cc9
1:M 22 Jul 2026 08:19:24.149 * <timeseries> Redis version found by RedisTimeSeries : 8.8.0 - oss
1:M 22 Jul 2026 08:19:24.149 * <timeseries> Registering configuration options: [
1:M 22 Jul 2026 08:19:24.149 * <timeseries>     { ts-compaction-policy   :             }
1:M 22 Jul 2026 08:19:24.149 * <timeseries>     { ts-num-threads         :           3 }
```


8. docker exec -it my_database sh

```
C:\Users\15L>docker exec -it my_database sh
# exit
```

9. stop my_database

```
C:\Users\15L>docker stop my_database
my_database
```

10. docker ps --all

```
C:\Users\15L>docker ps --all
CONTAINER ID   IMAGE                          COMMAND                  CREATED         STATUS                     PORTS
               NAMES
9b0e3e996f29   redis                          "docker-entrypoint.s…"   4 minutes ago   Exited (0) 23 seconds ago
               my_database
8ddb1e79e05f   ubuntu                         "/bin/bash"              19 hours ago    Exited (0) 19 hours ago
               distracted_ishizaka
298abd2b1a9d   c234a24851ba                   "docker-entrypoint.s…"   3 months ago    Exited (0) 3 months ago
               naughty_shirley
1a197c2bd0f5   c234a24851ba                   "docker-entrypoint.s…"   3 months ago    Exited (255) 19 hours ago   6379/tc
p              happy_galois
013ed608bab8   c234a24851ba                   "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago
               nifty_yonath
f9b579837c27   d1397258b209                   "docker-entrypoint.s…"   4 months ago    Exited (0) 4 months ago
               vigilant_knuth
06b8f6ece314   ca2b0f26964c                   "/bin/bash"              2 years ago     Exited (255) 4 months ago
               happy_cerf
7b8921bf4fb0   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (255) 3 months ago   6379/tc
p              jatinn
c782c1137761   node                           "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               jatingupta
ca1d141bfabd   node                           "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               reverent_williams
7638b03139a0   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               jatin
d53dc9583439   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               happy_engelbart
45c72eb69cd7   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               keen_blackburn
b9b0d800f902   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
               priceless_jackson
a91dfde56008   d1397258b209                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago
```

11. docker rm my_database docker rmi redis

```
C:\Users\15L>docker rmi redis
Untagged: redis:latest
Deleted: sha256:234c902a2db49461a129e2d4aeff85b28cf20187ed274a67f6e50995fa713c7b
```

docker

12. images

```
C:\Users\15L>docker images

IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
my-app:1.0           07c9f9065fb5       157MB         41.6MB
my-app:latest        e1b8c6a9dbf7       157MB         41.6MB
my-ubuntu-app:v1     9ddf795e7b17       157MB         41.6MB
nginx:latest         5a88c9c45479       241MB           66MB   U
ubuntu:latest        3131b4cc82a7       161MB         45.3MB   U
```

13. docker search nginx

```
C:\Users\15L>docker search nginx
NAME                                      DESCRIPTION                                     STARS     OFFICIAL
nginx                                     Official build of Nginx.                        21340     [OK]
nginx/nginx-ingress                       NGINX and  NGINX Plus Ingress Controllers fo…  122
nginx/nginx-prometheus-exporter           NGINX Prometheus Exporter for NGINX and NGIN…  52
nginx/nginx-ingress-operator              NGINX Ingress Operator for NGINX and NGINX P…  4
nginx/nginxaas-loadbalancer-kubernetes                                                    1
bitnamicharts/nginx                       Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx                              Nginx, a high-performance reverse proxy & we…  141
kasmweb/nginx                             An Nginx image based off nginx:alpine and in…  9
rancher/nginx                                                                             4
linuxserver/nginx                         An Nginx container, brought to you by LinuxS…  236
dtagdevsec/nginx                          T-Pot Nginx                                     0
vmware/nginx                                                                              3
paketobuildpacks/nginx                                                                    0
gluufederation/nginx                       A customized NGINX image containing a consu…  1
cleanstart/nginx                          Secure by Design, Built for Speed, Hardened …  0
activestate/nginx                         ActiveState's customizable, low-to-no vulner…  0
antrea/nginx                              Nginx server used for Antrea e2e testing        0
intel/nginx                                                                               0
docksal/nginx                             Nginx service image for Docksal                 1
geokrety/nginx                            Our customized nginx image                      0
circleci/nginx                            This image is for internal use                  2
ilios/nginx                               Nginx customized to run Ilios along with the…  0
corpusops/nginx                           https://github.com/corpusops/docker-images/     1
dockette/nginx                            Nginx SSL / HSTS / HTTP2                        3
wayofdev/nginx                            Nginx Docker image for PHP development. SSL-…  0
```

14. docker pull nginx

```
C:\Users\15L>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
```

15. docker run -d --name my_webserver nginx

```
C:\Users\15L>docker run -d --name my_webserver nginx
25c6bd3cddd652b31e0bd4e3a91c2213456ee9b6b13bba1b0207b0b38812906c
```

16. exec -it my_webserver sh

```
C:\Users\15L>docker exec -it my_webserver sh
# exit
```

17. docker stop my_webserver docker rm my_webserver

docker

```
C:\Users\15L>docker stop my_webserver
my_webserver

C:\Users\15L>docker rm my_webserver
my_webserver
```

18. docker rmi nginx docker images

```
C:\Users\15L>docker images

IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
my-app:1.0         07c9f9065fb5       157MB         41.6MB
my-app:latest      e1b8c6a9dbf7       157MB         41.6MB
my-ubuntu-app:v1   9ddf795e7b17       157MB         41.6MB
nginx:latest       5a88c9c45479       241MB           66MB   U
ubuntu:latest      3131b4cc82a7       161MB         45.3MB   U
```

19. docker pull mysql

```
C:\Users\15L>docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
81b3a6a3bce8: Pull complete
e0607e526570: Pull complete
5ac73e0e3f6a: Pull complete
c2fb254fc2f4: Pull complete
61d49498d3da: Pull complete
e21df1635f12: Pull complete
08b808cb6d40: Pull complete
bb49c3208fe7: Pull complete
1576763b528a: Pull complete
88745f4755f0: Pull complete
a701ef10ab46: Download complete
1f9f19898688: Download complete
Digest: sha256:8fdf311514c91fa5014e93e98e19d7f5d9eb162a462c13100c956dacc278ee21
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest
```

20. run -d --name lab_db -e MYSQL_ROOT_PASSWORD=secret mysql

```
C:\Users\15L>docker exec -it lab_db mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

docker

CREATE TABLE stud (roll_no INT PRIMARY KEY, name VARCHAR(50), marks INT););

```
mysql> use lab_workexp2;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-----------------------+
| Tables_in_lab_workexp2 |
+-----------------------+
| stud                  |
+-----------------------+
1 row in set (0.001 sec)

mysql> INSERT INTO stud (roll_no, name, marks)
    -> VALUES (1, 'Raj', 95);
Query OK, 1 row affected (0.017 sec)

mysql> select*from stud;
+---------+------+-------+
| roll_no | name | marks |
+---------+------+-------+
|       1 | Raj  |    95 |
+---------+------+-------+
1 row in set (0.000 sec)
```

Conclusion:

In this experiment, the foundational concepts of Docker and application containerization were successfully studied and practically applied. The theoretical framework established the role of Docker in providing standardized, isolated environments, while clearly distinguishing between static, immutable Docker images and active, writable Docker containers.

Through the hands-on implementation phase, a complete container lifecycle was executed using a standard database image (redis,nginx). The practical steps successfully demonstrated how to:

Search for and pull specific images from a registry.

Deploy an image into an active, isolated container running in the background.

Monitor, inspect, and access the internal shell of a live container.

Perform proper environment cleanup by stopping the container, removing it, and deleting the underlying image.

Ultimately, this experiment provided the working knowledge necessary to package, deploy, and manage isolated software environments consistently across different systems.